EXPECTED

Data Structures

&

Algorithms

Question Types for the Term End Examination

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Disclaimer

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"Homo in Luna ambulavit."
"Man has walked on the Moon."

Time: 3 hours Maximum Marks: 100

Answer Q1 and any four questions from Q2–Q7. Each question carries 20 marks. For revision, attempt every question and the alternative in Q6(a).

Write algorithms as numbered steps: Step 0: Start; Step 1, Step 2, ...; End. Show intermediate states. Use zero-based array indices and state any additional assumptions.

Q1. Answer briefly.

(a) An array stack has capacity 5 and is initially empty. Trace:
`Push(10), Push(20), Push(30), Peek(), Pop(),`
`Push(40), Push(50), Push(60), Push(70).`
Show the stack after every operation, returned values, and any overflow. [5]

(b) For $[4, 3, 1, 5, 2]$, show every pass of selection sort and insertion sort. State which standard implementation is stable. [5]

(c) Define the degree, depth and height of a tree node. Explain the difference between preorder, inorder and postorder traversal. State your height convention. [5]

(d) Choose and justify a suitable structure for (i) expected constant-time passenger lookup by ticket ID and (ii) repeatedly scheduling the job with the shortest running time. [5]

Q2.

(a) Explain O, Ω and Θ notation. Determine the tight time bound and auxiliary space of the following function, with reasons. Assume constant-time arithmetic. [10]

```
Analyze(A, n):
    s = 0
    for i = 0 to n-1:
        s = s + A[i]
    for i = 1 to n:
        for j = 1 to i:
            s = s + 1
    return s
```

(b) (i) Write a stack algorithm to check balanced brackets; trace it on `(a-c)*(b+c)`. [5]
(ii) Evaluate `9 3 4 * 8 + 4 / -`, showing the stack after each token and the order in which operands are popped. [5]

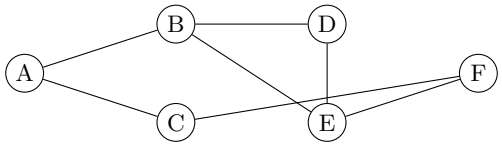
Q3.

(a) Starting with $10 \to 20 \to 30 \to 40 \to$ NULL, insert 25 after node 20, then delete the node containing 30. Draw both pointer changes. Write algorithms for insertion after a known node and deletion by value, including empty-list, head-node and not-found cases. [10]

(b) Construct a BST by inserting $50, 30, 70, 20, 40, 60, 80, 65$ in order. Draw the tree after each insertion. Explain the three deletion cases; delete 50 using its inorder successor and show every change. Give the final inorder traversal. [10]

Q4.

(a) Write BFS and DFS algorithms. Traverse the following undirected graph from A, considering neighbours alphabetically. Show the BFS queue and DFS active stack after each discovery, and give both orders. [5+5]



(b) Write merge-sort and merge algorithms. Sort $39, 9, 81, 45, 90, 27, 72, 18$, showing every split and merge. Explain the time and auxiliary-space complexities. [10]

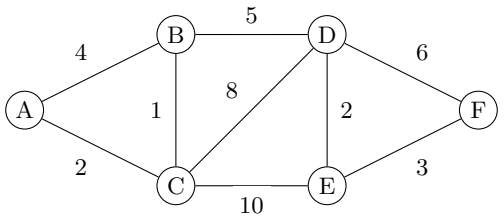
Q5.

(a) (i) Write an infix-to-postfix algorithm. Convert $((A+B)*C-D)/(E+F)$, showing the operator stack and output after each token. [7]
(ii) Convert `P Q R * + S -` to fully parenthesised infix using a stack. [3]

(b) Write the postorder traversal algorithm. Reconstruct the binary tree from the sequences below, showing every partition and node attachment. Give its preorder traversal. [10]
Inorder: $4, 8, 2, 5, 1, 6, 3, 7$.
Postorder: $8, 4, 5, 2, 6, 7, 3, 1$.

Q6.

(a) For the weighted, undirected graph below, define a minimum spanning tree. Construct one using Prim's algorithm from A, then verify it using Kruskal's algorithm. Show each chosen edge, any rejected edge inspected, and the total cost. [5+5]



OR

Write Dijkstra's algorithm. Using the same graph with A as source, show every settling and relaxation step, the final distance to each vertex, and the shortest routes to E and F. Break distance ties alphabetically. [10]

(b) Explain linear probing. Insert $27, 37, 47, 57, 15, 25, 35$ into a table of size 10 using $h(k) = k \bmod 10$. Show every probe, including wraparound, and the final table. Find the load factor and explain two disadvantages of linear probing. [10]

Q7.

(a) Write circular-queue enqueue and dequeue algorithms, including full and empty tests. Use capacity 5 and initially $front = rear = -1$. Enqueue 10,20,30,40; dequeue twice; then enqueue 50,60,70,80. Show the array, front, rear and logical FIFO order after every operation; identify any overflow. Give two applications. [10]

(b) Write Huffman's algorithm. For frequencies $A : 5$, $B : 9$, $C : 12$, $D : 13$, $E : 16$, $F : 45$, show the priority queue after every merge and draw every newly formed subtree. Assign 0 to left edges and 1 to right edges. Give all codes and the total encoded bit count; compare it with fixed-length coding, excluding codebook overhead. [10]